

Hash

	Slot 0	Slot 1
0	Asus	Apple
1		
2	Canon	
3	Dell	
4	Epson	Ericsson
...		
25	Zyxel	

Loading density

$$\alpha = n/sb$$

- n = number of pairs in the table
- s = number of slots per bucket
- b = number of buckets

• insert (Cisco, 1000) will

trigger Collision

↳ the bucket is not empty

• insert (Acer, 2000) will

trigger Overflow

↳ the bucket is full

hashing methods

✱ The more random, the better

folding at the boundaries 123|456|789 $\Rightarrow (123+654+789) \% m$

Overflow handling

linear probing

- tombstone
- Set the deleted position to "deleted" instead of "empty"
 - Record the number of tombstones to check if $h[i]$ is full

backward shifting Shift the rear elements to the front when deleting
↳ only for linear probing

quadratic probing

$$k, k \pm 1^2, k \pm 2^2, k \pm 3^2, \dots \pmod{b}$$

it should be the prime number of the form $4j+3$

bloom filter

1	1	0	1	1	0	0
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$$h_1(\text{"cat"}) = 1, h_1(\text{"dog"}) = 0$$

$$h_2(\text{"cat"}) = 3, h_2(\text{"dog"}) = 4$$

- If all the result of h_k is true, then the element may be in $ht[]$
- Otherwise, the element should not be in $ht[]$

ex:

$$h_1(\text{"lion"}) = 1$$

$$h_2(\text{"lion"}) = 4$$

Design

Given the filter size m and the number of elements u
False positive rate is minimized if $h = (m \ln 2) / u$

- If h is too small, probability of different keys having same signature is high
- If h is too large, the bloom filter fills with ones quickly